TEXORA · LECTURE SHEET

Magnetism

Magnetic induction and the fields of currents, force and torque on conductors and loops, the Hall effect, Faraday's and Lenz's laws, self and mutual induction, magnetic materials and hysteresis, and L–C oscillations as electromagnetic simple harmonic motion.

PHY-1203 · PHYSICS-II · TOPIC 2 · LEVEL 1 TERM 2

1. Magnetic induction — the field of a current

The **magnetic induction** B (tesla) is defined by the force on a moving charge, F = qv × B. Currents create B (Biot–Savart / Ampère):

SOURCE	B
Long straight wire, distance r	$\mu_0 I/(2\pi r)$, circles round the wire
Centre of a circular loop (N turns, radius R)	$\mu_0 NI/(2R)$
Inside a long solenoid (n turns/m)	$\mu_0 nI$, uniform; ≈ 0 outside
Inside a toroid	$\mu_0 NI/(2\pi r)$

WORKED EXAMPLES

Wire carrying 10 A, at r = 5 cm: B = $2 \times 10^{-7} \times 10/0.05$ = **4×10^{-5} T** (about Earth's field — a memorable scale).
Loop: N = 50, I = 2 A, R = 10 cm: B = $(4\pi \times 10^{-7} \times 100)/(0.2)$ = **6.28×10^{-4} T** at the centre.

2. Force on a conductor; torque on a loop

A conductor of length L carrying I in field B feels **F = BIL sinθ** (θ between current and field), direction by Fleming's left-hand rule — the working principle of motors, meters and loudspeakers. A rectangular loop (N turns, area A) in a field feels no net force but a **torque**

$$\tau = NIBA \sin\theta$$

(θ between B and the loop's normal), maximum when the loop's plane lies along B. The moving-coil galvanometer balances this torque against a spring (with radial field to keep $\tau \propto I$), making deflection

proportional to current.

WORKED EXAMPLES

B = 0.5 T, I = 10 A, L = 0.20 m, θ = 90°: F = 0.5 × 10 × 0.2 = **1.0 N**.

Loop N = 100, I = 2 A, B = 0.1 T, A = 0.010 m² at θ = 90°: τ = 100 × 2 × 0.1 × 0.010 = **0.20 N·m**.

3. The Hall effect

Pass current through a slab in a perpendicular B: the magnetic force herds the charge carriers to one face until the transverse (Hall) field balances it. The **Hall voltage** across the slab's width is

$$V_H = IB/(nqt)$$

(t = thickness along B, n = carrier density, q = carrier charge). Its sign reveals the carriers' sign (electrons in most metals; holes in some semiconductors) and its size measures n or B — Hall probes are standard magnetic-field meters. For copper ($n \approx 8.5 \times 10^{28}$ m^{-3}) the voltage is tiny (fractions of a μV), which is why semiconductor Hall elements dominate practice.

4. Electromagnetic induction — Faraday and Lenz

Faraday's law: the induced emf equals the rate of change of flux linkage:

$$e = -N\, d\Phi/dt, \quad \Phi = BA\cos\theta$$

Lenz's law is the minus sign: the induced current flows so as to oppose the change producing it — energy conservation in disguise (the work you do against the opposition is the electrical energy gained). **Self-induction:** a coil's own changing current induces e = −L dI/dt; L (henry) is the self-inductance, and the coil stores energy ½LI². **Mutual induction:** a changing current in one coil induces $e_2 = -M\, dI_1/dt$ in a neighbour; M depends on geometry and coupling.

WORKED EXAMPLES

Coil of 200 turns, flux changing by 0.05 Wb in 0.10 s: |e| = 200 × 0.05/0.10 = **100 V**.

L = 0.5 H with current changing at 10 A s^{-1}: |e| = 0.5 × 10 = **5 V**.

Energy in a 2 H inductor carrying 3 A: ½ × 2 × 9 = **9 J**.

5. Classification of magnetic materials

CLASS	BEHAVIOUR	EXAMPLES / NOTES

Diamagnetic	Weakly repelled; μ_r slightly < 1; induced atomic moments oppose B	Copper, water, bismuth — universal but feeble
Paramagnetic	Weakly attracted; μ_r slightly > 1; permanent atomic moments align partially	Aluminium, platinum; effect fades as 1/T (Curie law)
Ferromagnetic	Strongly attracted; μ_r up to thousands; domains align; hysteresis; loses ferromagnetism above the Curie temperature	Iron, steel, nickel, cobalt — motors, transformers, memory

6. Hysteresis

Magnetising a ferromagnet traces B against H: the curve climbs, saturates, and on reducing H the magnetisation **lags** — at H = 0 a remanent induction B_r (retentivity) remains; a reverse field −H_c (coercivity) is needed to erase it; the loop closes only after a full cycle. The **area of the B–H loop** is the energy dissipated as heat per unit volume per cycle — why transformer cores use soft iron (narrow loop, low loss) and permanent magnets use hard steel (wide loop, high B_r and H_c). Repeated cycling through the loop is also how a magnet is demagnetised (cycle with shrinking amplitude).

7. L–C oscillations — electromagnetic SHM

A charged capacitor discharging through an inductor oscillates: charge and current trade energy between the capacitor's electric field ($Q^2/2C$) and the inductor's magnetic field ($\frac{1}{2}LI^2$), with total conserved (resistance absent):

$$Q = Q_0 \cos(\omega t), \quad \omega = 1/\sqrt{(LC)}, \quad f = 1/(2\pi\sqrt{(LC)})$$

The analogy with simple harmonic motion is exact and worth tabulating:

MECHANICAL SHM	ELECTRICAL L–C
Displacement x	Charge Q
Velocity v	Current I
Mass m (inertia)	Inductance L
Spring constant k	1/C
$\frac{1}{2}kx^2$ (spring energy)	$Q^2/2C$ (capacitor energy)
$\frac{1}{2}mv^2$ (kinetic)	$\frac{1}{2}LI^2$ (inductor energy)

WORKED EXAMPLE

L = 2 mH, C = 50 pF: $LC = 10^{-13}$; $f = 1/(2\pi\sqrt{(10^{-13})}) = 1/(2\pi \times 3.16 \times 10^{-7}) \approx$ **503 kHz** — squarely in the radio band, which is no coincidence: this circuit is the ancestor of every tuner (Topic 1's resonance with R → 0).

8. Key points

- B of wires, loops and solenoids all come from $\mu_0 I \times$ geometry; learn the three standard results cold.
- F = BIL sinθ drives motors; τ = NIBA sinθ drives meters.
- Hall voltage: V_H = IB/(nqt); sign → carrier sign, size → n or B.
- Faraday: e = −N dΦ/dt; Lenz's minus is energy conservation. Self: e = −L dI/dt, energy ½LI².
- Dia/para/ferro: feeble-opposing, feeble-aligning, domain-driven strong — only ferromagnets show hysteresis.
- Hysteresis loop area = energy lost per cycle per volume; soft magnetic materials minimise it, hard ones maximise remanence.
- L–C circuit = SHM with L as mass and 1/C as spring constant; $f = 1/(2\pi\sqrt{(LC)})$.

9. Practice

Answer in your own words; brief model answers follow.

1. A 50 cm wire carrying 5 A lies at 30° to a field of 0.8 T. Find the force on it.
2. A 500-turn coil links flux that changes by 0.02 Wb in 0.04 s. Find the induced emf.
3. For a copper strip ($n = 8.5 \times 10^{28}$ m^{-3}) of thickness 0.10 mm carrying 2 A in a field of 0.5 T, compute the Hall voltage and comment on the size.
4. Why is the core of a transformer laminated, and why soft iron rather than steel? Answer through hysteresis.
5. An L–C circuit has L = 1 mH and C = 100 pF. Find its natural frequency, and the maximum current if the capacitor was charged to 100 V (resistance negligible).
6. State Lenz's law and use it to find the direction of the induced current when a magnet's north pole is pushed toward a coil's face. What happens on withdrawal?

Answer key

1. F = BIL sinθ = 0.8 × 5 × 0.50 × sin 30° = 0.8 × 5 × 0.50 × 0.5 = 1.0 N.
2. |e| = N ΔΦ/Δt = 500 × 0.02/0.04 = 250 V.

3. $V_H = IB/(net) = (2 \times 0.5)/(8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 1.0 \times 10^{-4}) = 1/(1.36 \times 10^{6}) \approx 7.4 \times 10^{-7}$ V — under a microvolt. Metals' huge n makes their Hall voltages minute; semiconductors (n smaller by ~10^6) give usable millivolts, which is why Hall devices are semiconductor-based.
4. Each A.C. cycle drags the core round its hysteresis loop, dissipating the loop's area as heat. Soft iron's loop is narrow (small area, low coercivity), minimising that loss; steel's wide loop would waste far more. Laminating the core (insulated sheets) separately suppresses eddy currents, the other core loss — the two measures attack the two losses.
5. $f = 1/(2\pi\sqrt{(10^{-3} \times 10^{-10})}) = 1/(2\pi \times 3.16 \times 10^{-7}) \approx 503$ kHz. Energy: $Q_0 = CV = 10^{-10} \times 100 = 10^{-8}$ C; $U = Q_0^2/(2C) = 10^{-16}/(2 \times 10^{-10}) = 5 \times 10^{-7}$ J. At peak current all energy is magnetic: $\frac{1}{2}LI_0^2 = 5 \times 10^{-7} \rightarrow I_0 = \sqrt{(2 \times 5 \times 10^{-7}/10^{-3})} = \sqrt{(10^{-3})} \approx 0.032$ A = 32 mA.
6. Lenz: the induced current opposes the change producing it. Approaching north pole increases the flux into the coil's face, so the induced current makes that face a north pole (repelling the magnet) — anticlockwise as seen by the approaching magnet. On withdrawal the flux falls, the current reverses, making the face a south pole that pulls the magnet back. Both times you must do work against the opposition — that work is the induced electrical energy.
